13. Design and configure a network (wired and wireless LANs) with multiple subnets using required network devices. Configure at least three of the following services in the network- TELNET, SSH, FTP server, Web server, File server, DHCP server and DNS server.

Aim

Design and configure a network with multiple subnets with wired and wireless LANs using required network devices. Configure the following services in the network - TELNET, SSH, FTPserver, Web server, File server, DHCP server and DNS server.

Theory

Subnet

A subnet is a logical partition of an IP network into multiple, smaller network segments. It is typically used to subdivide large networks into smaller, more efficient subnetworks. The internet is composed of many networks that are run by many organizations. In turn, each organization's network can be composed of many smaller networks, or subnets. Each subnet allows its connected devices to communicate with each other, and routers are used to communicate between subnets. The size of a subnet depends on the connectivity requirements and the network technology employed. A point-to-point subnet allows two devices to connect, while a data center subnet might be designed to connect many more devices.

Configuring the Services

The following shows how the different services can be configured in an Ubuntu PC:

Telnet

Telnet is a user command and an underlying TCP/IP protocol for accessing remote computers. Through Telnet, an administrator or another user can access someone else's computer remotely. On the Web, HTTP and FTP protocols allow you to request specific files from remote computers, but not to actually be logged on as a user of that computer. With Telnet, you log on as a
regular user with whatever privileges you may have been granted to the specific application and data on that computer. Telnet is most likely to be used by program developers and anyone who has a
need to use specific applications or data located at a particular host computer.

- Install Telnet
  sudo apt install telnet xinetd
- Edit /etc/inetd.conf with root permission, add this line:
  telnet stream tcp nowait telnetd /usr/sbin/tcpd /usr/sbin/in.telnetd
- Edit /etc/xinetd.conf, copy the following configuration:

```
# Simple configuration file for xinetd
#
# Some defaults, and include /etc/xinetd.d/
defaults
  {
  # Please note that you need a log_type line to be able to use
```

# log_on_success and log_on_failiure The default is the following:
# log_type = SYSLOG daemon info
instances = 60
log_type = SYSLOG authpriv
log_on_success = HOST PID
log_on_failiure = HOST
cps = 25 30
}

- Change telnet port by using the following command in the terminal:
  telnet 23/ tcp
- Then restart the service and check runnig status:
  sudo systemctl restart inetd
  sudo systemctl status inetd

```
hishamalip@savage:~$ sudo systemctl restart inetd
hishamalip@savage:~$ sudo systemctl status inetd
● xinetd.service - LSB: Starts or stops the xinetd daemon.
     Loaded: loaded (/etc/init.d/xinetd; generated)
     Active: active (running) since Thu 2020-04-30 17:49:41 IST; 2s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 29339 ExecStart=/etc/init.d/xinetd start (code=exited, status=0/SUCCESS)
      Tasks: 1 (limit: 8940)
     Memory: 1.9M
     CGroup: /system.slice/xinetd.service
             └─29367 /usr/sbin/xinetd -pidfile /run/xinetd.pid -stayalive -inetd_compat -inetd_ipv6

Apr 30 17:49:41 savage xinetd[29367]: Reading included configuration file: /etc/xinetd.d/discard-udp
Apr 30 17:49:41 savage xinetd[29367]: Reading included configuration file: /etc/xinetd.d/echo [file=
Apr 30 17:49:41 savage xinetd[29367]: Reading included configuration file: /etc/xinetd.d/echo-udp [f
```

- Login to telnet using:
  telnet ip_address

```
hishamalip@savage:/etc/xinetd.d$ telnet 192.168.43.178
Trying 192.168.43.178...
Connected to 192.168.43.178.
Escape character is '^]'.
Ubuntu 20.04 LTS
savage login: s1767
Password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage


19 updates can be installed immediately.
19 of these updates are security updates.
To see these additional updates run: apt list --upgradable

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Thu Apr 30 00:38:06 IST 2020 on tty4
s1767@savage:~$ ls
ftp
s1767@savage:~$ cd ftp
s1767@savage:~/ftp$
```

SSH

The SSH protocol (also referred to as Secure Shell) is a method for secure remote login from one computer to another. It provides several alternative options for strong authentication, and it protects the communications security and integrity with strong encryption. It is a secure alternative to the non-protected login protocols (such as telnet, rlogin) and insecure file transfer methods (such as FTP).

Take the following steps to configure SSH:

- Install SSH
  - sudo apt install
  - (Installing the client can be done by replacing openssh-server by openssh-client)
- Configure SSH:
  - sudo nano /etc/ssh/sshd_config
  - Then make the changes you want to make
- Restart SSH
  - sudo systemctl restart ssh
- Verify the runnig status of SSH
  - sudo systemctl status ssh

If configuration is successful,You should see something like Active: active (running)

```
hishamalip@savage:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2020-04-30 13:56:30 IST; 1h 42min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 1171 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1231 (sshd)
      Tasks: 1 (limit: 8940)
     Memory: 2.5M
     CGroup: /system.slice/ssh.service
             └─1231 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Apr 30 13:56:27 savage systemd[1]: Starting OpenBSD Secure Shell server...
Apr 30 13:56:30 savage sshd[1231]: Server listening on 0.0.0.0 port 22.
Apr 30 13:56:30 savage sshd[1231]: Server listening on :: port 22.
Apr 30 13:56:30 savage systemd[1]: Started OpenBSD Secure Shell server.
hishamalip@savage:~$
```

- We can login to the SSH server from an SSH client.
  - ssh username@ip_address

```
hishamalip@savage:~$ ssh s1767@192.168.43.178
s1767@192.168.43.178's password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage


14 updates can be installed immediately.
14 of these updates are security updates.
To see these additional updates run: apt list --upgradable

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Wed Apr 29 17:05:13 2020 from 192.168.43.178
s1767@savage:~$ ls /
bin    dev   lib    libx32      mnt   root  snap  tmp
boot   etc   lib32  lost+found  opt   run   srv   usr
cdrom  home  lib64  media       proc  sbin  sys   var
s1767@savage:~$ logout
Connection to 192.168.43.178 closed.
hishamalip@savage:~$
```

FTP Server

File Transfer Protocol (FTP) is the commonly used protocol for exchanging files over the Internet. FTP uses the Internet's TCP/IP protocols to enable data transfer. FTP uses a client-server architecture, often secured with SSL/TLS. FTP promotes sharing of files via remote computers with reliable and efficient data transfer. FTP uses a client-server architecture. Users provide authentication using a sign-in protocol, usually a username and password, however some FTP servers may be configured to accept anonymous FTP logins where you don't need to identify yourself before accessing files. Most often, FTP is secured with SSL/TLS.

The following steps show setting up an FTP server on the computer:

- Install FTP daemon
  sudo apt install vsftpd
- Configuring FTP can be done by editing the following file:
  sudo nano / etc / vsftpd . Conf
- Restart the service and check the status of vsftp
  sudo systemctl restart vsftpd
  sudo systemctl status vsftpd

```
hishamalip@savage:~$ sudo systemctl status vsftpd.service
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2020-04-30 16:07:35 IST; 2min 4s ago
    Process: 13778 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 13783 (vsftpd)
      Tasks: 1 (limit: 8940)
     Memory: 692.0K
     CGroup: /system.slice/vsftpd.service
             └─13783 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 30 16:07:35 savage systemd[1]: Starting vsftpd FTP server...
Apr 30 16:07:35 savage systemd[1]: Started vsftpd FTP server.
hishamalip@savage:~$
```

- Connecting to ftp server

  ftp -p [IP_Address]

```
hishamalip@savage:~/Desktop/vsftpd_test$ ftp -p 192.168.43.178
Connected to 192.168.43.178.
220 (vsFTPd 3.0.3)
Name (192.168.43.178:hishamalip): s1767
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,43,178,167,76).
150 Here comes the directory listing.
drwxr-xr-x    2 1001     1001         4096 Apr 29 18:04 files
226 Directory send OK.
ftp> cd files
250 Directory successfully changed.
ftp> ls
227 Entering Passive Mode (192,168,43,178,174,38).
150 Here comes the directory listing.
-rw-r--r--    1 0        0              17 Apr 29 17:54 test.txt
226 Directory send OK.
ftp> get test.txt
local: test.txt remote: test.txt
227 Entering Passive Mode (192,168,43,178,178,183).
150 Opening BINARY mode data connection for test.txt (17 bytes).
226 Transfer complete.
17 bytes received in 0.00 secs (190.8226 kB/s)
ftp> 221 Goodbye.
hishamalip@savage:~/Desktop/vsftpd_test$ ls
test.txt
hishamalip@savage:~/Desktop/vsftpd_test$ cat test.txt
vsftpf test file
hishamalip@savage:~/Desktop/vsftpd_test$
```

Web Server

A Web server is a program that uses HTTP (Hypertext Transfer Protocol) to serve the files that form Web pages to users, in response to their requests, which are for warded by their computers' HTTP clients. Dedicated computers and appliances may be referred to as Web servers as well. The process is an example of the client/server model. All computers that host Web sites must have Web server programs. Leading Web servers include Apache (the most widely installed Web server), Microsoft's Internet Information Server (IIS) and nginx (pronounced engine X) from NGNIX. Other Web servers include Novell's NetWare server, Google Web Server (GWS) and IBM's family of Domino servers.

A web server can be hosted on the localhost of the PC by following the following steps:

- Install Apache server

  sudo apt install apache2
- Check the webserver status

  sudo systemctl status apache2

```
hishamalip@savage:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset:
     Active: active (running) since Thu 2020-04-30 13:56:34 IST; 4h 59min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1307 (apache2)
      Tasks: 55 (limit: 8940)
     Memory: 9.2M
     CGroup: /system.slice/apache2.service
             ├─1307 /usr/sbin/apache2 -k start
             ├─1308 /usr/sbin/apache2 -k start
             └─1309 /usr/sbin/apache2 -k start
```

- Hosting a website:
  - Create the directory for your_domain:
    sudo mkdir /var/www/[domain-name]
  - Assign ownership of the directory:
    sudo chown -R $USER:$USER /var/www/[domain-name]
  - Create webpages inside the folder [domain-name] to access it
  - Make a new virtual host file at /etc/apache2/sites-available/[domain-name.conf] and paste the following example to configure it
    sudo nano /etc/apache2/sites-available/your_domain.conf

```
<VirtualHost *:80>
        ServerAdmin webmaster@localhost
        ServerName your_domain
        ServerAlias your_domain
        DocumentRoot /var/www/your_domain
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

  - Enabling the configuration file by using the command a2ensite
    sudo a2ensite [domain-name].conf
  - Disable the default site defined in 000-default.conf:
    sudo a2dissite 000-default.conf
  - Test for configuration errors:
    sudo apache2ctl configtest

If the configuration is successful then corresponding output will displayed

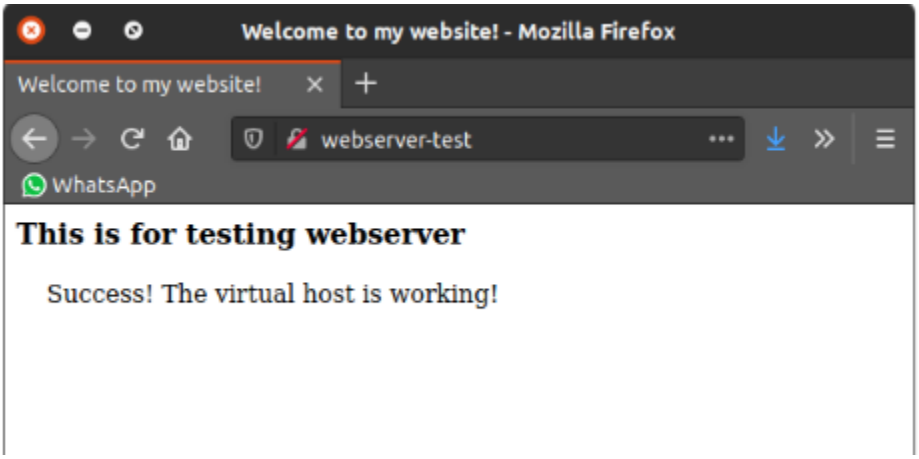
```
hishamalip@savage:~$ sudo apache2ctl configtest
Syntax OK
hishamalip@savage:~$
```

- By editing /etc/hosts file, we can give the domain name for the website

```
  GNU nano 4.8                    /etc/hosts
127.0.0.1       localhost
127.0.1.1       savage
127.0.1.1       webserver-test

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

- The configuration the server is in the file: /etc/apache2/apache2.conf
- Restart the server by the command:
  sudo systemctl restart apache2
- By vising the [domain-name] we can access the website that we created



File Server

In the client/server model, a file server is a computer responsible for the central storage and management of data files so that other computers on the same network can access the files. A file server allows users to share information over a network without having to physically transfer files by floppy diskette or some other external storage device. Any computer can be configured to be a host and act as a file server. In its simplest form, a file server may be an ordinary PC that handles requests for files and sends them over the network. In a more sophisticated network, a file server might be a dedicated network-attached storage (NAS) device that also serves as a remote hard disk drive for other computers, allowing anyone on the network to store files on it as if to their own hard drive.

The following steps can be followed to setup a file server:

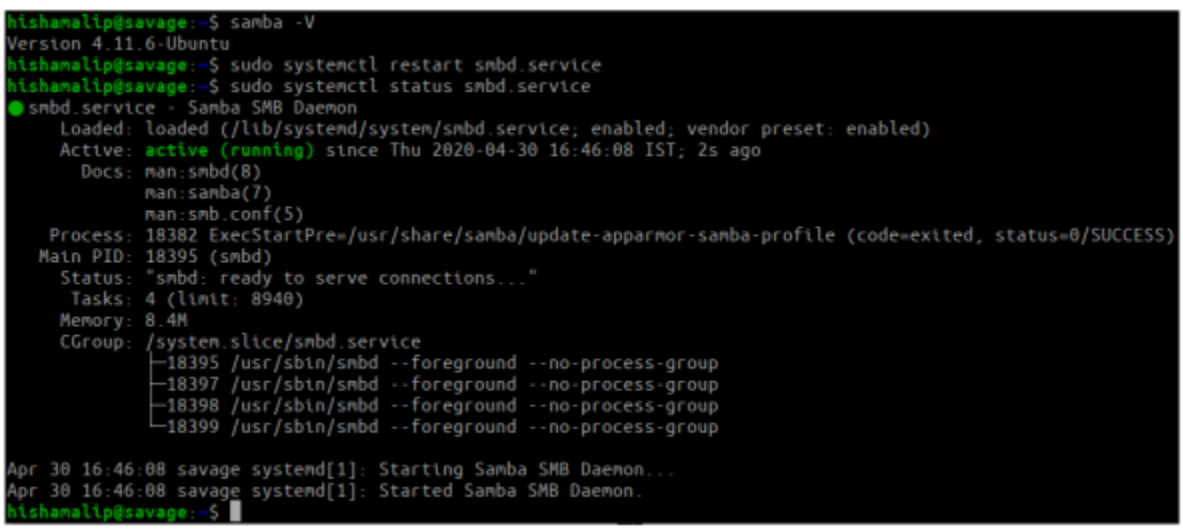
- Installing Samba File Server:
  sudo apt install samba
- Configuring the file server by editing /etc/samba/smb.conf
  First, edit the following key/value pairs in the [global] section of /etc/samba/smb.conf:

workgroup = EXAMPLE
...
security = user

Create a new section at the bottom of the file, or uncomment one of the examples, for the directory to be

shared:
[share]
comment = UbuntuFileServerShare
path = /srv/samba/share
browsable = yes
guest ok = y e s
read only = no
create mask = 0755

- Make a directory for hosting files and setting permission for the directory:
  sudo mkdir -p /srv/samba/share/
  sudo chown nobody:nogroup /srv/samba/share/
- Restart Samba service and checking status:
  sudo systemctl restart smbd.service nmbd.service
- We can connect to samba server using sama



DHCP Server

DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to dynamically assign an Internet Protocol (IP) address to any device, or node, on a network so they can communicate using IP. DHCP automates and centrally manages these configurations rather than requiring network administrators to manually assign IP addresses to all network devices. DHCP can be implemented on small local networks as well as large enterprise networks. DHCP will assign new IP addresses in each location when devices are moved from place to place, which means network administrators do not have to manually initially configure each device with a valid IP address or reconfigure the device with a new IP address if it moves to a new

location on the network. Versions of DHCP are available for use in Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6).

The following steps shows how DHCP server can be run:

- Install DHCP server:

  sudo apt install isc-dhcp-server

- Configure DHCP server, the config file is /etc/dhcp/dhcpd.conf :

```
# Sample /etc/dhcpd.conf
# ( add your comments here)
default-lease-time 600;
max-lease-time 7200;
option subnet-mask 255.255.255.0;
option broadcast-address 192.168.1.255
option routers 192.168.1.254;
option domain-name-sever 192.168.1.1 , 192.168.1.2
option domain-name “mydomain.example”;
subnet 192.168.1.0 netmask 255.255.255.0 {
range 192.168.1.100;
range 192.168.1.150 192.168.1.200;
}
```

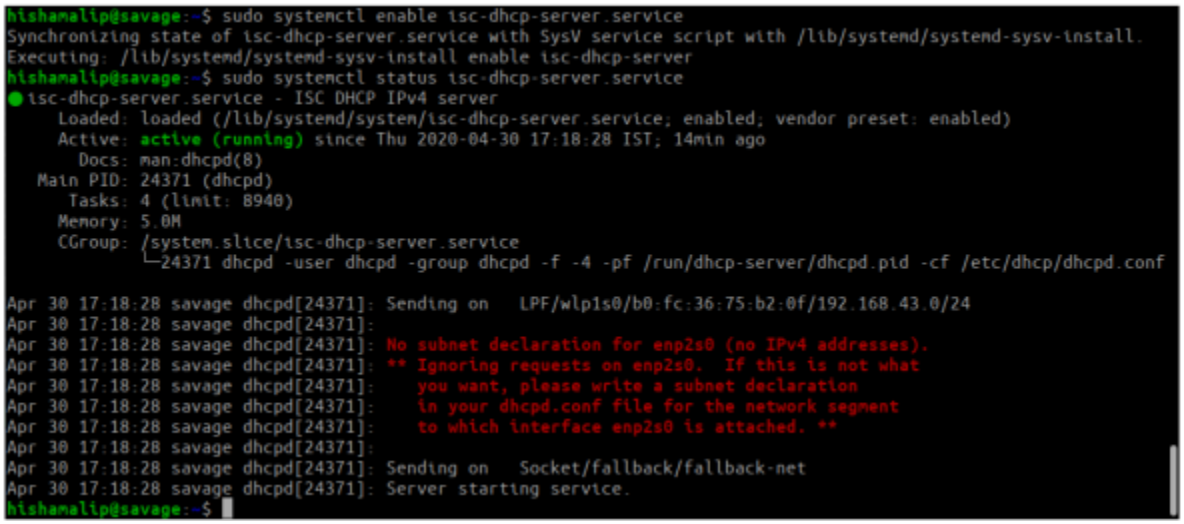
- Starting and stopping services can be achieved using:

  sudo service isc-dhcp-server restart
  sudo service isc-dhcp-server start
  sudo service isc-dhcp-server stop

  After editing configuration files, we have to restart the service

- Enable and check the status of DHCP Server

  sudo systemctl enable isc-dhcp-server.service
  sudo systemctl status isc-dhcp-server.service



```
hishamalip@savage:~$ sudo systemctl enable isc-dhcp-server.service
Synchronizing state of isc-dhcp-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
hishamalip@savage:~$ sudo systemctl status isc-dhcp-server.service
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2020-04-30 17:18:28 IST; 14min ago
       Docs: man:dhcpd(8)
   Main PID: 24371 (dhcpd)
      Tasks: 4 (limit: 8940)
     Memory: 5.0M
     CGroup: /system.slice/isc-dhcp-server.service
             └─24371 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf

Apr 30 17:18:28 savage dhcpd[24371]: Sending on   LPF/wlp1s0/b0:fc:36:75:b2:0f/192.168.43.0/24
Apr 30 17:18:28 savage dhcpd[24371]:
Apr 30 17:18:28 savage dhcpd[24371]: No subnet declaration for enp2s0 (no IPv4 addresses).
Apr 30 17:18:28 savage dhcpd[24371]: ** Ignoring requests on enp2s0.  If this is not what
Apr 30 17:18:28 savage dhcpd[24371]:    you want, please write a subnet declaration
Apr 30 17:18:28 savage dhcpd[24371]:    in your dhcpd.conf file for the network segment
Apr 30 17:18:28 savage dhcpd[24371]:    to which interface enp2s0 is attached. **
Apr 30 17:18:28 savage dhcpd[24371]:
Apr 30 17:18:28 savage dhcpd[24371]: Sending on   Socket/fallback/fallback-net
Apr 30 17:18:28 savage dhcpd[24371]: Server starting service.
hishamalip@savage:~$
```

DNS Server

The Domain Name Systems (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DN S translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in Ipv4), or more complex newer alphanumeric IP address such as 2400:cb00:2048:1::c629:d7a2 (in Ipv6).

The following steps show the setup:

- Installing:
  sudo apt install bind9
- The configuration is in the /etc/bind folder
- Setting as a catching name server by editing the file /etc/bind/named.conf.options:
  forwarders {
  1.2.3.4; # replace with the ip address
  5.6.7.8; # of the name servers
  };
- BIND9 can be configured with the primary and the secondary master as a custom DNS server to access all the subnets.

```
options {
        directory "/var/cache/bind";

        // If there is a firewall between you and nameservers you want
        // to talk to, you may need to fix the firewall to allow multiple
        // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

        // If your ISP provided one or more IP addresses for stable
        // nameservers, you probably want to use them as forwarders.
        // Uncomment the following block, and insert the addresses replacing
        // the all-0's placeholder.

         forwarders {
                8.8.8.8;
                8.8.4.4;
        };

        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================
        dnssec-validation auto;

        listen-on-v6 { any; };
};
```

- Restarting bind9:
  sudo systemctl restart bind9

- You can use the dig command to test DNS services

```
hishamalip@savage:~$ dig ubuntu.com

; <<>> DiG 9.16.1-Ubuntu <<>> ubuntu.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 4371
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;ubuntu.com.                    IN      A

;; ANSWER SECTION:
ubuntu.com.             202     IN      A       91.189.88.181
ubuntu.com.             202     IN      A       91.189.91.44
ubuntu.com.             202     IN      A       91.189.91.45
ubuntu.com.             202     IN      A       91.189.88.180

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Thu Apr 30 20:18:07 IST 2020
;; MSG SIZE  rcvd: 103

hishamalip@savage:~$
```

Default Samba Configuration

The main Samba configuration file is called /etc/samba/smb.conf. Before we start editing smb.conf config file, let's make a backup of the original configuration file and extract a current configuration relevant lines to a new smb.conf file:
user@fisat:~# cp /etc/samba/smb.conf /etc/samba/smb.conf_backup
user@fisat:~# grep -v -E "^#|^;" /etc/samba/smb.conf_backup | grep . > /etc/samba/smb.conf

Now the /etc/samba/smb.conf should now contain:
user@fisat:~# cat /etc/samba/smb.conf
[global]
workgroup = WORKGROUP
dns proxy = no
log file = /var/log/samba/log.%m
max log size = 1000
syslog = 0
panic action = /usr/share/samba/panic-action %d
server role = standalone server
passdb backend = tdbsam
obey pam restrictions = yes
unix password sync = yes
passwd program = /usr/bin/passwd %u
passwd chat = *Enter\snew\s*\spassword:* %n\n *Retype\snew\s*\spassword:* %n\n *password\supdated\ssuccessfully* .

pam password change = yes
map to guest = bad
user usershare allow guests = yes
[homes] comment = Home Directories browseable = no read only = yes create mask = 0700
directory mask = 0700
valid users = %S
[printers]
comment = All Printers
browseable = no
path = /var/spool/samba
printable = yes
guest ok = no
read only = yes
create mask = 0700
[print$]
comment = Printer Drivers

path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no

● Restart your samba server and use SMB client to confirm all exported samba groups:
user@fisat:~# systemctl
restart smbd user@fisat:~# smbclient -L localhost
WARNING: The "syslog" option is deprecated
Enter root's password:
Domain=[WORKGROUP] OS=[Windows 6.1] Server=[Samba 4.5.8-Debian]
Sharename Type Comment
print$ Disk Printer Drivers
IPC$ IPC IPC Service
Domain=[WORKGROUP] OS=[Windows 6.1] Server=[Samba 4.5.8-Debian]

Server Comment --------- ------- LASERPRINTER LINUXCONFIG Samba 4.5.8-Debian
Workgroup Master --------- ------- WORKGROUP LINUXCONFIG

Result:

For accessing the different nodes in the subnet, TELNET, SSH, FTP server, Web server, File server, DHCP server and DNS server have been configured and runs successfully in an Ubuntu 20.04 LTS .